

1. Administrative & Study Identification

Full Study Title: A Modular Phase I/IIa, Open-label Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Efficacy of Ascending Doses of AZD5335 Monotherapy and in Combination With Anti-cancer Agents in Participants With Solid Tumors **Short Title/Acronym:** FONTANA **Protocol Number:** D8990C00001 **NCT Number:** NCT05797168 **Trial Status:** RECRUITING **Sponsor Name:** AstraZeneca AB **Investigational Product:** AZD5335 (Folate Receptor α (FR α)-targeted Antibody-Drug Conjugate) +/- Combinations (e.g., Saruparib/AZD5305, Bevacizumab, Carboplatin, AZD9574) **Phase of Development:** Phase I/IIa **Medical Monitor:** AstraZeneca Clinical Operations Lead (Niloufar Davoudi)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of ascending doses of AZD5335 monotherapy and in combination with other anti-cancer agents, and to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To characterize the pharmacokinetics (PK) profile, evaluate preliminary anti-tumor efficacy (ORR, DoR, DCR), and observe the disease control rate in patients with solid tumors.

2.2 Background & Rationale Certain advanced solid tumors, notably ovarian cancer, lung adenocarcinoma (LUAC), and endometrial cancer, exhibit high levels of Folate Receptor alpha (FR α) expression. AZD5335 is an antibody-drug conjugate (ADC) designed to bind to FR α and deliver a highly potent topoisomerase 1 inhibitor payload directly into tumor cells. This targeted delivery aims to provide competitive efficacy with a manageable safety profile in both high and low FR α -expressing tumors, meeting a high unmet need in patients who have progressed on standard therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Modular assignment, Dose-escalation [Part A] & Dose-optimization [Part B]) **Blinding:** Open-label **Randomization:** N/A (Sequential assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Estimated 506 participants **Number of Sites:** Multi-center (International)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of signing informed consent. 2. Histologically or cytologically confirmed advanced solid tumors (e.g., ovarian cancer, LUAC, endometrial cancer) with a life expectancy of ≥ 12 weeks. 3. Participants must have received prior adequate therapy in accordance with local practice for their tumor type and stage of disease. 4. Measurable disease per RECIST v1.1. 5. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 6. Adequate organ and marrow function. 7. Participants with past or resolved HBV/HCV or well-controlled HIV (undetectable viral RNA, CD4+ count $\geq 350/\text{mm}^3$) are eligible under specific conditions. 8. Highly effective contraception usage (men must use a male condom post-screening through 5 months following the last dose of study intervention; women must use highly effective birth control for 8 months post-dose).

4.2 Exclusion Criteria 1. History of another primary malignancy (except for malignancies treated with curative intent and no known active disease for ≥ 2 years, adequately treated non-melanoma skin cancer, or carcinoma in situ). 2. Spinal cord compression, history of leptomeningeal carcinomatosis, or active brain metastases (unless asymptomatic, stable, and not requiring continuous corticosteroids > 10 mg prednisone/day for at least 4 weeks prior to first dose). 3. Unresolved toxicities of Grade ≥ 2 (NCI CTCAE v5.0) from prior therapy (excluding vitiligo, alopecia, and controlled endocrine disorders). 4. History of interstitial lung disease (ILD) or pneumonitis that required oral/IV steroids or supplemental oxygen. 5. Cardiac criteria including uncontrolled hypertension, symptomatic heart failure (NYHA class ≥ 2), prior/current cardiomyopathy, severe uncorrected valvular heart disease, acute coronary syndrome/MI/PCI/CABG within 6 months prior to screening, or a mean resting QTcF > 470 msec.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ascending doses of AZD5335 administered on Day 1 of each 3-week (21-day) cycle. Combinations (Module 2+) incorporate specific dosing of targeted agents like the PARP inhibitor Saruparib (AZD5305) or Carboplatin. **Route of Administration:** Intravenous (IV) infusion for AZD5335; Oral or IV for combination partners. **Duration of Treatment:** Cycles of 3 weeks continuing until unacceptable toxicity, disease progression, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms or treatment-related adverse events (e.g., anti-emetics for nausea/vomiting). **Prohibited:** Concurrent investigational agents, live vaccines within 30 days prior to first dose and during treatment, or overlapping anti-cancer therapies outside of the protocol-defined combinations.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (Archival/Baseline), ECG, Baseline Scans
Baseline	Day 0	Final eligibility confirmation, First Dose of Study Intervention
Interim	Every 3 Weeks	Treatment administration, Safety Labs, PK Blood Draws, Vital Signs, Efficacy Scans (e.g., Q6W-Q8W)
End of Study	Follow-up	Off-treatment safety follow-up, Survival tracking, Final scan evaluations

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Assessment of safety and tolerability defined by: 1. Number of participants with incidence of adverse events and serious adverse events, including changes from baseline in laboratory parameters, vital signs, ECGs, and physical examination. 2. Number of participants with Dose-Limiting Toxicities (DLTs) to establish the Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose. **Measurement Method:** Clinical observation, vital signs, ECGs, and central/local laboratory analyses graded per NCI CTCAE v5.0.

7.2 Secondary & Exploratory Endpoints 1. Preliminary efficacy defined by Objective Response Rate (ORR), Duration of Response (DoR), and Disease Control Rate (DCR) per RECIST v1.1. 2. Pharmacokinetics (PK): Evaluation of terminal elimination half-life, plasma concentrations of AZD5335, total antibody, and unconjugated payload.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via physical exams, laboratory tests, and ECG monitoring. Common anticipated AEs include nausea, fatigue, neutropenia, vomiting, and anemia. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. Close monitoring for adverse events of special interest such as ILD/pneumonitis. **Data Monitoring Committee (DMC):** An internal Safety Review Committee (SRC) oversees dose-escalation decisions during Part A of the modular study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Safety Set includes all patients who receive at least one dose of the investigational product. The Efficacy Analysis Set includes patients with measurable baseline disease who receive ≥ 1 dose. **Sample Size Justification:** Target $N=506$ accommodates the modular, multi-cohort design necessary for finding optimal dosing in both monotherapy and distinct combination arms. **Handling of Missing Data:** For ORR/efficacy endpoints, patients without post-baseline scan assessments are generally considered non-responders per RECIST v1.1.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standardized site visits and remote data verification by AstraZeneca to review source documentation, safety events, and protocol compliance. **Audit Trail:** Robust tracking of eCRF completion, data clarification queries, and secure maintenance of optional genetic research consent logs.

11. Study Identifiers & Governance

Primary ID: {{D8990C00001}} **Registry Number:** {{NCT05797168}}
Trial Status: {{RECRUITING}} **Sponsor/Lead Organization:** {{AstraZeneca AB}} **Study Title:** {{Phase I/IIa Study of AZD5335 as Monotherapy and Combination Therapy in Participants With Solid Tumors}} **Official Regulatory Title:** {{A Modular Phase I/IIa, Open-label Study to Assess the Safety, Tolerability, Pharmacokinetics and Preliminary Efficacy of Ascending Doses of AZD5335 Monotherapy and in Combination With Anti-cancer Agents in Participants With Solid Tumors}}

12. Clinical Framework (Oncology Specific)

Primary Condition: {{Advanced Solid Tumors (Ovarian Cancer, Lung Adenocarcinoma, Endometrial Cancer)}} **Diagnosis Threshold:** {{Histologically or cytologically confirmed advanced malignancy}} **Medical Methodology:** {{Folate Receptor α (FR α)-Targeted Antibody-Drug Conjugate (ADC) Therapy}} **Optimization Target:** {{Determination of Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D)}}

13. Study Procedures & Methodology

Management Pathway: {{Modular Dose Escalation (Part A) and Dose Optimization (Part B)}} **Education Protocol (HCP):** {{Protocol training, DLT criteria tracking, and ILD/pneumonitis safety monitoring}} **Education Protocol (Patient):** {{Informed consent process, contraceptive counseling, and symptom reporting}} **Quality Audit Mechanism:** {{Centralized medical data review and SRC dose-escalation gating}}

14. Observation & Follow-Up Schedule

Total Duration: {{Estimated 55 Months (June 2023 - January 2028)}} **Onsite Visit Cadence:** {{Every 3 Weeks (Day 1 of each 21-day cycle)}} **Remote Monitoring Frequency:** {{Intermittent follow-up for post-treatment survival and safety checks}} **Checklist Review Interval:** {{Efficacy assessments via CT/MRI every 6 to 8 weeks depending on the cycle phase}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				